

# Computational Estimation Techniques for Initial Value Problems of Ordinary Differential Equations

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## 1 Introduction

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Ordinary differential equations come into play in the heart of the universe. From most commonly in mechanics, to fluids and sound waves, there is perhaps hardly a situation in the universe where ODEs don't come into play. Hence, the study of them and how to obtain solutions of them, are of fundamental significance. This paper explores some recursive and computational methods to estimate the solutions to ordinary differential equations with a given initial condition, which are collectively called initial value problems, with the aim of exposing the reader to these computational methods in a concise way that can be further used for application using computer-aided engineering and such. We will build from the ground up, introducing what ordinary differential equations and their solutions are, to the methods developed by researchers over time.

## 2 Defining an ODE

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For a function  $y(x)$  of an independent variable  $x$ , a differential equation is one where atleast a derivative of the function with respect to the independent variable is present. In this sense, even the equation  $dy/dx = 0$  is by definition a differential equation. There can be higher order derivatives, more independent variables or functions involved as well. For example, take two functions of the independent variable time ( $t$ ), velocity ( $v$ ) and displacement ( $s$ ).

An ordinary differential equation is a special case where there are only ordinary derivatives involved ( $\frac{d^i y}{dx^i}$ ) and there is only one independent variable, and no product of the independent variable and the derivative of the dependent variables w.r.t the independent variable occurs. In this sense, equations such as  $\frac{dy}{dx} + e^y = x$  are considered ordinary, while something like  $\frac{d^2 y}{dx^2} + \frac{d^5 y}{dt^2} = 10t$  cannot be considered ordinary as it contains more than one independent variable,  $t$  and  $x$ .

### 3 Defining a solution to an ODE

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Consider an ordinary differential equation of degree  $n \in \mathbf{Z}$ , of the form

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 y + a_0 = f(x)$$

where  $a_i \in R$ . We can also write this in the form of

$$F\left(x, y, \frac{d^n y}{dx^n}, \frac{d^{n-1} y}{dx^{n-1}}, \dots, \frac{dy}{dx}\right) = 0$$

A solution for the above ODE, of the form  $g(x)$  is defined such that when we substitute  $y = g(x)$ , the given equation is satisfied.

A 'general' solution of an ODE is one where there are undetermined constants at play. For example, take the ODE  $\frac{dy}{dx} = \frac{\sin x}{y}$ . This is very straightforward to solve. At the end of the integration, we get the solution  $y = \sqrt{-2\cos(x) + c}$ , where  $c$  is the constant of integration. This is the general solution of the above case.

A particular solution is one where atleast one constant has been identified. Consider the case  $\frac{dy}{dx} = e^x$ . This is very straightforward to solve as well. At the end of the integration, we get the general solution  $y = e^x + c$ . Putting  $c = 4$ , we get a particular solution  $y_1 = e^x + 2$ .

In it's essence, obtaining a solution is just being able to express the given ODE into an equation without any derivatives involved, just the dependent and independent variables in some combination, be it algebraic, trigonometric, exponential, or even in the complex plane.

### 4 Initial Value Problems (IVPs)

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A particular solution to an  $n^{th}$  order ordinary differential equation is only possible to obtain when we have information about where the curve of the  $n-1^{th}$  derivative of the dependent variable w.r.t the independent variable passes through. So, for an ODE of degree 1, the like of the examples we've seen above, to obtain a particular solution, we need to know a particular point that the  $1-1 = 0^{th}$  derivative, or the curve of  $y$  itself passes through. So it is necessary to have a condition  $y(x_0) = y_0$  along with the first order ODE.

This condition or set of conditions, or the information about where the function itself and/or the derivatives pass through, for any  $n^{th}$  order ODE is called formally as an initial value condition. This condition or set of conditions coupled along with the ordinary differential equation are collectively called an initial value problem.

Now that we have established the groundwork, it is obvious that obtaining solutions to these IVPs is significant. From here, we will overview some of the methods developed that give close approximates to the solutions of the IVPs, or even just the ODEs by themselves without any given initial value condition. Some methods require exclusively a computer to carry out iterative tasks to get to an approximation.

## 5 Computational methods for solutions

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The Runge-Kutta methods are a series of methods for the estimation of non-linear differential equations developed by German mathematicians Carl Runge and Wilhelm Kutta. Some of the simplest ones in the family of these methods are the forward and backward euler method.

### 5.1 Forward Euler Method (RK1)

This method exclusively works on approximating the solution to a first-order ODE. Consider the initial value problem

$$\begin{aligned}\frac{dy}{dx} &= f(x, y) \\ y(x_0) &= y_0\end{aligned}$$

This is termed as an explicit method, as it solves explicitly for  $\frac{dy}{dx}$  on one side. The backward euler method uses the same core idea, but with difference of the points used. Here we focus on the forward method. The point on the cartesian plane  $(x_0, y_0)$  can be considered our starting point. By definition, the derivative of the function at a point is the slope of the curve at that point. The curve of a smooth function can be imagined to be just a collection of infinitely small straight lines joined end to end, with their slopes matching the slope of the curve at any point. This is the basis of the forward euler method.

Choosing a step-size  $h \in \mathbb{R}$  where  $h > 0$ , we can iteratively plot points along the plane and join them, matching the 'shape' of the curve to obtain. The smaller the step size, the smoother the curve. For the first point from the starting point, we can write

$$y_1 = y_0 + hf(x_0, y_0)$$

Moving the step  $h$  along the x axis, such that  $x_1 = x_0 + h$ , we obtain the second point,

$$y_2 = y_1 + hf(x_1, y_1)$$

In general, the formula for any one point can be diluted into

$$y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})$$

Using a computer, we can iteratively carry out this process for smaller step sizes, yielding a high-accuracy polygonal curve estimation of the particular solution curve. While this method exclusively works on first-order ODEs, any  $n^{th}$  order ODE can be decomposed into first-order ODEs and further solved. However, this paper will not explore that.

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### 5.2 Direction Fields

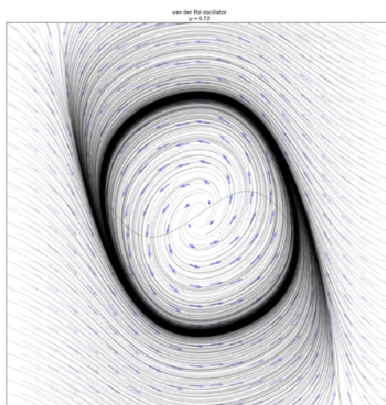
Direction fields, also called slope fields, are a visualization of the solution curves by intuition. As we know, the derivative of the dependent variable w.r.t the independent variable at any point is the slope of it's curve at that point. Again, smooth curves are just infinitesimally small line segments joined end to end, where the midpoint of each line segment can be considered as the point where we're approximating the function, and the slope given by the derivative of the function at that point.

So, when we are explicitly given the derivative of the dependent variable at any point in the form of  $\frac{dy}{dx} = g(x, y)$  we can visualize the 'flow' of the function in general by iteratively plotting short straight line segments along the entire cartesian plane, where the slope of the line segment is equal to the value of the derivative at that point.

While direction fields don't give particular solutions to an IVP or an ODE on it's own, they can be extremely useful in studying the behavior of the dependent variable near trivial points. They're often used in electrical engineering, where an obtained ODE is sometimes too hard, or even considered impossible to solve. A good example of this would be the second order **Van der Pol equation of electronics**:

$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x = 0$$

where  $x$  is the position coordinate of the oscillator,  $t$  is time, and  $\mu$  is a damping coefficient. In the given phase plot of the equation, the blue arrows are direction fields.




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### 5.3 Picard's Method

Picard's method is yet another iterative method for obtaining the solution to a first order IVP. This uses integrals and stems from the **fundamental theorem of calculus** to obtain solutions iteratively, and therefore is suitable for when the established integrals are solvable. For the given IVP

$$\begin{aligned}\frac{dy}{dx} &= f(x, y) \\ y(x_0) &= y_0\end{aligned}$$

By the fundamental theorem of calculus, for an independent variable  $t$  with dependent variables  $G(t)$  and  $g(t, G(t))$  where  $\frac{dG}{dt} = g(t)$ ,

$$\int_a^b g(t)dt = G(b) - G(a)$$

Using this, we can write the above established formulation, using  $b = x_0$  and an arbitrary point  $x_1 > x_0$  as  $a = x_1$ , as

$$\begin{aligned}
\int_{x_0}^{x_1} f(x, y) dx &= y(x_1) - y(x_0) \\
\implies \int_{x_0}^{x_1} f(x, y) dx &= y(x_1) - y_0 \\
\implies y(x_1) &= \int_{x_0}^{x_1} f(x, y) dx + y_0 \\
\implies y(x) &= \int_{x_0}^x f(t, y) dt + y_0 \\
\implies \phi(x) &= \int_{x_0}^x f(t, y) dt + \phi_0
\end{aligned}$$

So, we obtain a general formula to be used iteratively

$$\phi_n(x) = \phi_0 + \int_{x_0}^x f(t, \phi_n(t)) dt$$

For the use of integrals, this method is only applicable when the parent integral of  $\int f(t, \phi_n(t)) dt$  is solvable.

## 5.4 Phase Line Method

An autonomous ODE is an explicit ODE that depends only on the dependent variable. In other words, it can be written in the form of

$$\frac{dy}{dx} = f(x)$$

only. Observing this, we can note down a property, that the slope of the solution curve at any point only depends only the x coordinate. In other words, the slope of the solution curves along any horizontal line would be the same. We can say the solution curves are just copies of each other, differing by their intercepts, or in other words displaced horizontally by some amount.

From here, we use phase lines to show the regions where the solution increases, and where it decreases. By simple calculus, we can find regions where

- $f(x) > 0$ , or where the solution increases
- $f(x) < 0$ , or where the solution decreases
- $f(x) = 0$ , or where the solution neither increases nor decreases. These are called **equilibrium points**.

Selectively for point of equilibrium  $x_0$ , we can show the dependence of the solution from the left hand side and the right hand side of the region. The cases can be

- **Unstable Equilibrium:**  $f(x) > 0$  for  $x < x_0$  and  $f(x) < 0$  for  $x > x_0$
- **Stable Equilibrium:**  $f(x) < 0$  for  $x < x_0$  and  $f(x) > 0$  for  $x > x_0$

## 5.5 Taylor Series Method

The Taylor series is an excellent approximation of any function  $f(x)$ , which only requires the value of derivatives of the function at a fixed point  $x_0$ . The more values we have, the better the approximation is.

For any function  $f(x)$ , the Taylor series expansion can be written as

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots + \frac{f^n(x_0)}{n!}(x - x_0)^n$$

In summation form, it can be written as

$$f(x) = \lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{f^{(i)}(x_0)(x - x_0)^i}{i!}$$

This gives close approximations to the actual function. So say we have an explicitly defined IVP,

$$\frac{dy}{dx} = f(x, y)$$

$$f(x_0) = y_0$$

Our goal is to find the solution  $\phi(x)$  such that

$$\frac{d\phi}{dx} = f(x, y)$$

$$\phi(x_0) = y_0$$

So, similarly using the Taylor series, we can expand  $\phi(x)$  as

$$\phi(x) = \phi(x_0) + \phi'(x_0)(x - x_0) + \frac{\phi''(x_0)}{2!}(x - x_0)^2 + \dots + \frac{\phi^n(x_0)}{n!}(x - x_0)^n$$

Now, we will apply some fundamental rules of calculus to find any  $\phi^i(x_0)$ .

$$\phi(x_0) = y_0$$

$$y' = f(x, y)$$

$$\implies \phi'(x_0) = f(x_0, y_0)$$

For the second derivative, that is,  $\phi''(x_0)$ ,

$$\phi' = f(x, y)$$

$$\implies \frac{d\phi}{dx} = f(x, y)$$

$$\implies \frac{d^2\phi}{dx^2} = \frac{df(x, y)}{dx}$$

The total derivative of this multi-variable function  $f(x, y)$  is given by

$$df = \frac{\delta f}{\delta x}dx + \frac{\delta f}{\delta y}dy$$

$$\Rightarrow \frac{df}{dx} = \frac{\delta f}{\delta x} + \frac{\delta f}{\delta y} \frac{dy}{dx}$$

Using this result, we can substitute it into the above equation, further getting the complete expression for the second derivative,

$$\begin{aligned} \frac{d^2\phi}{dx^2} &= \frac{\delta f}{dx} + \frac{\delta f}{\delta y} \frac{dy}{dx} \\ \Rightarrow \frac{d^2\phi}{dx^2} &= \frac{\delta f}{dx} + \frac{\delta f}{\delta y} f(x, y) \end{aligned}$$

Similarly, we can carry out this process to obtain the third derivative, and from there the fourth derivative and so on. The higher we go, the more accurate our solution gets. This method is especially useful for when the ordinary and partial derivatives can be easily calculated and substituted for the value.

## 6 Conclusion

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These are some of the computational estimation techniques of solutions to both IVPs and standalone ordinary differential equations, as well as autonomous equations.

Not every method is suitable for every case, each case of equations has its own best method of approximating solutions. In the physical world, establishing the mathematical model first of what we are studying, then obtaining test data for initial value conditions is the workflow to studying physical systems.